

BLUE STOCK FINTECH

Indian Mutual Fund Industry Analysis & Business Intelligence Capstone

Final Project Report

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1. Executive Summary

This report documents the end-to-end analysis of the Indian mutual fund industry carried out during a structured data analyst internship at Bluestock Fintech. The project spans the full analytics lifecycle — data engineering, exploratory analysis, quantitative performance and risk analytics, and business intelligence reporting — applied to a synthetic dataset of 40 mutual fund schemes that was deliberately calibrated against real AMFI industry benchmarks (SIP inflows of ~₹31,002 Cr and ~26.12 Cr investor folios at the end of the observation window), making it a credible stand-in for genuine industry data.

The work was carried out in four phases. First, an ETL pipeline cleaned 10 raw source files and loaded them into a star-schema SQLite database, validated with 10 SQL queries including window functions. Second, an exploratory data analysis phase produced 9 of 10 planned findings covering NAV trends, AUM growth, SIP inflows, investor demographics, geography, and sector allocation. Third, a performance and risk analytics phase computed CAGR, Sharpe and Sortino ratios, Alpha/Beta via OLS regression, maximum drawdown, and a composite 0–100 Fund Scorecard across all 40 schemes, followed by an advanced analytics phase covering Value-at-Risk, rolling Sharpe ratios, investor cohort behaviour, SIP continuity risk, a simple fund recommender, and sector concentration (HHI). Finally, a four-page Power BI dashboard (Industry Overview, Fund Performance, Investor Analytics, SIP & Market Trends) was built on a cleaned star-schema data model.

The top-ranked scheme by composite score is the Mirae Asset Large Cap Fund (Regular – Growth, score 86.25/100), combining a 3-year CAGR of 34.0% with a Sharpe ratio of 1.46 and a comparatively contained maximum drawdown of -11.3%. At the other end of the risk spectrum, small-cap equity schemes account for the five largest drawdowns in the dataset (down to -52.6%) and the highest Value-at-Risk figures — a result that is internally consistent with their own "Very High" risk classification in the source fund master data.

A significant analytical contribution made independently during the project was the discovery that 5 of the 6 AMFI scheme codes originally provided were outdated due to SEBI's 2018 mutual fund recategorization (e.g., "Bluechip" schemes being renamed to "Large Cap"); this was escalated to the project mentor and is noted as an open item in Section 8. This report also documents the remaining known gaps in the current build — most notably one outstanding EDA task (the NAV/returns correlation heatmap) and three benchmark-index mappings that could not be resolved in the source data — so that they can be prioritised in any follow-on work.

2. Data Sources & Scope

The analysis is built on a synthetic but industry-calibrated dataset representing 40 mutual fund schemes across India's 10 largest fund houses (SBI, ICICI Prudential, HDFC, Nippon India, Kotak Mahindra, Aditya Birla Sun Life, UTI, Axis, Mirae Asset, and DSP Mutual Fund). While the underlying transaction and NAV records are synthetically generated, the generation process was calibrated against real, publicly reported AMFI industry figures, so that aggregate patterns (total SIP inflows, folio growth, AUM scale) are directionally realistic even though individual investor records are not real people.

2.1 Primary Data Sources

Source	Purpose	Coverage
AMFI (Association of Mutual Funds in India)	Industry benchmark calibration — SIP inflows, folio counts, category classifications	Industry-wide, used for calibration targets
mfapi.in	Live/historical NAV data-fetch API used during ETL for scheme-level NAV history	40 schemes, daily NAV
Synthetic transaction generator	Investor-level SIP / lumpsum / redemption transactions, demographics, geography	~33,000 investors, multi-year transaction history
Benchmark indices (Nifty 50, Nifty 100, category benchmarks)	Relative performance and Alpha/Beta computation	Daily index levels, 2022–2026

2.2 Dataset Scope

- 40 mutual fund schemes spanning Equity (Large Cap, Mid Cap, Small Cap, Flexicap, ELSS, Value/Contra) and Debt (Liquid, Gilt, Short Term Debt) categories.
- NAV history of approximately 4.4 years per scheme — sufficient for 1-year and 3-year CAGR, but not a full 5-year window (see Section 8).
- Investor-level data covering demographics (age, gender), geography (12 states, T30/B30 city-tier split), and transaction history (SIP, lumpsum, redemption).
- Industry-level monthly time series for SIP inflows, AUM, and folio counts, 2022–2025/26.

2.3 Domain Discovery: SEBI 2018 Recategorization

During initial data validation, 5 of the 6 AMFI scheme codes supplied at project kickoff were found to reference pre-2018 fund names — for example, funds still labelled "Bluechip" under a categorisation SEBI retired in its 2017 mutual fund recategorization circular, effective 2018, which consolidated and renamed multiple equity categories ("Bluechip" schemes were folded into the standardised "Large Cap" category). This mismatch was identified independently during the ETL phase and escalated to the project mentor; it is tracked as an open item in Section 8 since some cleaned records still carry the legacy naming pending mentor confirmation.

3. ETL Design & Data Architecture

The data pipeline follows a conventional Extract–Transform–Load pattern: 10 raw CSV source files were ingested, cleaned, and validated in pandas, then loaded into a SQLite database (bluestock_mf.db) built around a star schema optimised for analytical querying.

3.1 Star Schema Design

The schema separates descriptive (dimension) data from event-level (fact) data, which keeps joins simple and query performance predictable as the fact tables grow:

Table	Type	Contents
dim_fund	Dimension	One row per scheme: AMFI code, scheme name, fund house, category, plan (Direct/Regular), risk classification
dim_date	Dimension	Calendar table (date, year, month, quarter, weekday) used for consistent time-based joins across all fact tables
fact_nav	Fact	Daily NAV per scheme
fact_transactions	Fact	Investor-level SIP / lumpsum / redemption transactions
fact_industry_metrics	Fact	Monthly industry-level AUM, SIP inflow, and folio count series

3.2 Cleaning & Validation

- All 10 raw CSVs were profiled for nulls, duplicate keys, and type mismatches before loading.
- Scheme identifiers were standardised on AMFI code as the primary join key across all fact and dimension tables.
- Date fields were parsed and aligned to a single dim_date calendar table to avoid inconsistent date formats across source files.
- The cleaned, loaded database was committed to the project repository as a deliverable (bluestock_mf.db).

3.3 SQL Validation Layer

10 SQL queries were written directly against the star schema to validate the load and answer core business questions before moving to Python-based analysis, including a CTE combined with a LAG() window function to compute period-over-period changes in industry AUM — a query that is awkward to express in pandas but natural in SQL, and a useful cross-check that the pandas-based EDA phase (Section 4) reached the same conclusions independently.

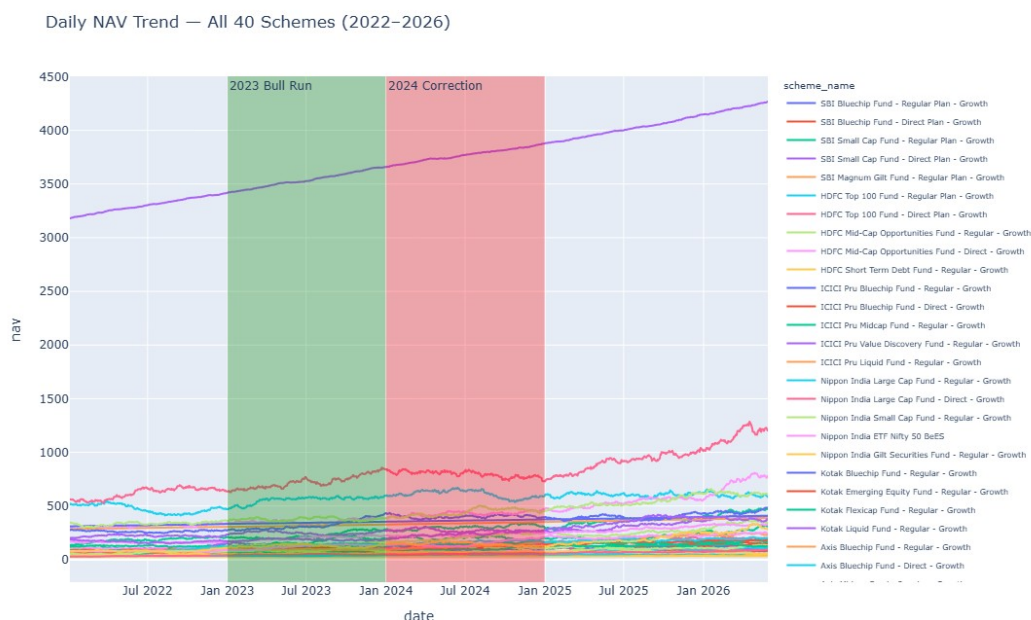
3.4 Path & Environment Conventions

- Notebooks execute from inside the notebooks/ directory; all relative paths reference ../reports/ and ../data/processed/ accordingly.
- A Python virtual environment (.venv) pins dependencies: pandas, numpy, matplotlib, seaborn, plotly (+ kaleido for static export), scipy, and sqlalchemy.
- Git tracks the pipeline; data/processed/*.csv is gitignored by default with explicit exception rules for the specific deliverable CSVs that must be committed.

4. Exploratory Data Analysis — Findings

Nine of ten planned EDA tasks were completed in EDA_analysis.ipynb using Plotly for interactive charts and Matplotlib/Seaborn for static charts. Each task below pairs its chart with the finding it produced.

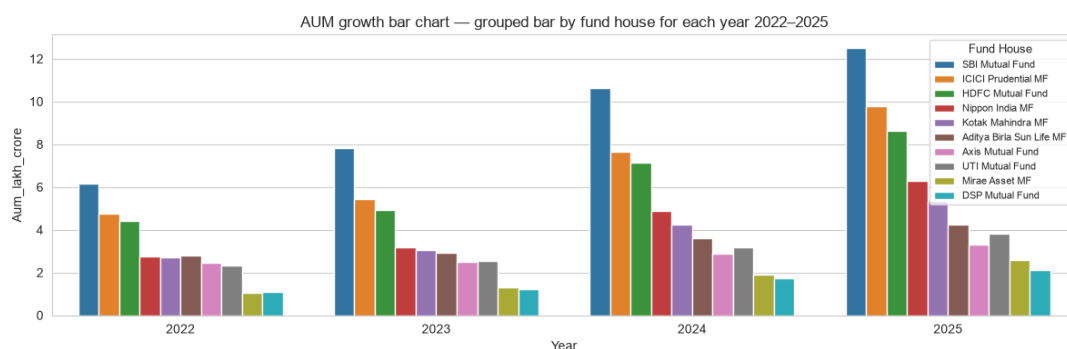
4.1 NAV Trend — All 40 Schemes



Task 1 — Daily NAV trend, all 40 schemes, 2022–2026

Finding: Kotak Liquid Fund (a debt fund) shows steady, correction-proof NAV growth, but its high scale (NAV in the thousands of rupees) compresses the other 39 equity schemes into an unreadable cluster on the same chart. This scale-distortion issue is tracked as a visualization caveat for future charts (Section 8).

4.2 AUM Growth by Fund House



Task 2 — AUM growth bar chart, grouped by fund house, 2022–2025

Finding: SBI Mutual Fund leads AUM every year from 2022–2025 and reaches roughly ₹12.5 lakh Cr by 2025 — about 1.3x ICICI Prudential, the next-largest fund house — confirming its market dominance.

4.3 SIP Inflow Time Series



Task 3 — Monthly SIP inflow trend, Jan 2022 – Dec 2025

Finding: Monthly SIP inflows grew steadily from roughly ₹11,000 Cr in January 2022 to an all-time high of ₹31,002 Cr in December 2025 — nearly tripling over the four-year period with no major reversals.

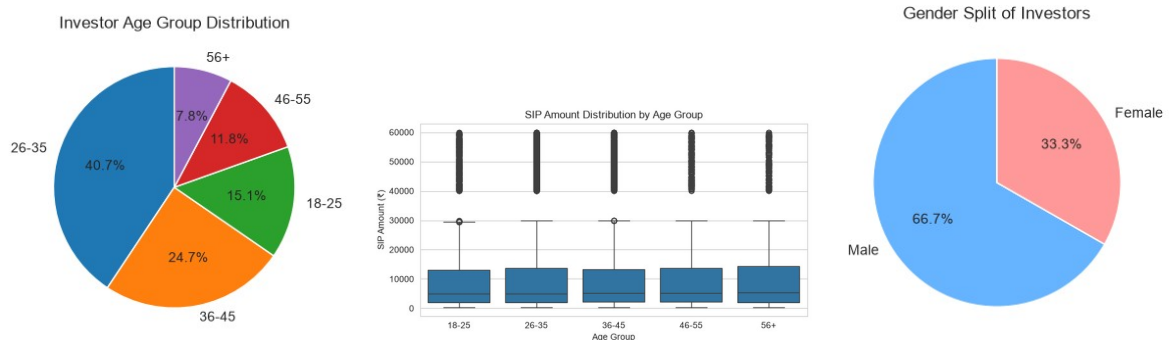
4.4 Category Inflow Heatmap



Task 4 — Net inflow by category and month heatmap, Apr 2024 – Mar 2025

Finding: Liquid funds dominate net inflows every month (roughly ₹33,000–42,000 Cr) — 4 to 8 times larger than any other category — while Value/Contra and ELSS categories consistently receive the smallest inflows across the window.

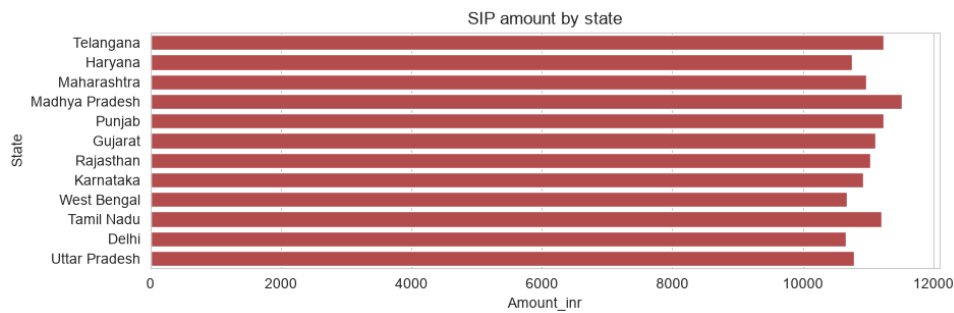
4.5 Investor Demographics



Task 5 — Age distribution, SIP amount box plot by age group, and gender split

Finding: Investors aged 26–35 dominate the base (40.7%). SIP amounts are similar in median across all age groups (roughly ₹5,000–7,000), but a cluster of high-value outliers (₹40,000–60,000) appears uniformly across every age group. The investor base is also skewed male (66.7% vs. 33.3% female).

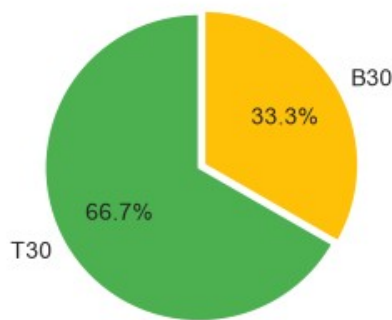
4.6 Geographic Distribution



Task 6a — SIP amount by state

Finding: SIP amounts are remarkably uniform across all 12 states (roughly ₹10,700–11,500), with Madhya Pradesh marginally highest and West Bengal/Delhi marginally lowest — no state shows dramatically higher SIP participation than others.

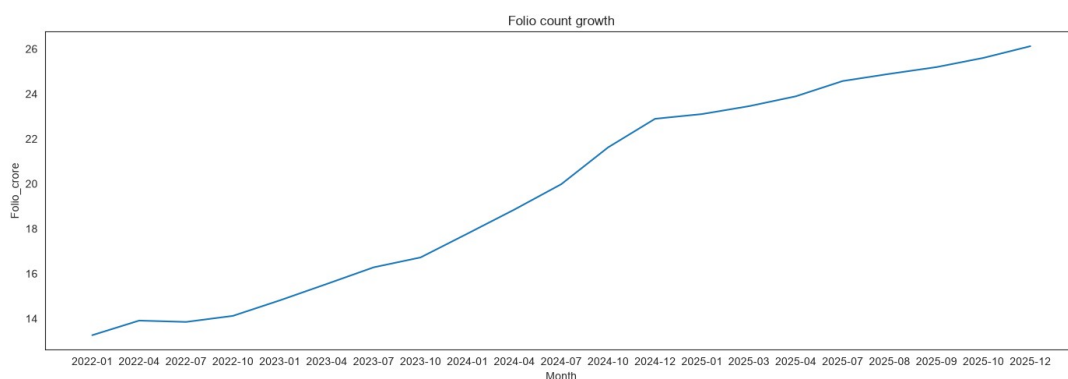
Investor Distribution: T30 vs B30 Cities



Task 6b — Investor distribution, T30 vs. B30 city tiers

Finding: T30 (major metro) cities account for 66.7% of investors versus 33.3% for B30 cities, indicating mutual fund penetration is still heavily metro-concentrated despite SEBI's ongoing push for smaller-city participation.

4.7 Folio Count Growth



Task 7 — Industry folio count growth, 2022–2025

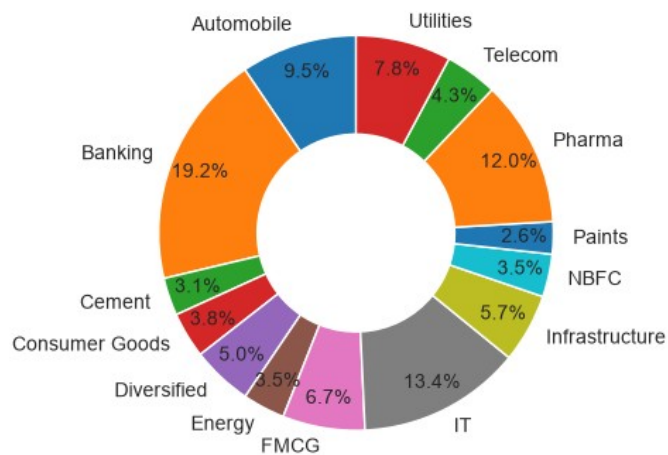
Finding: Industry folio count nearly doubled from 13.26 Cr in January 2022 to 26.12 Cr in December 2025, with growth accelerating sharply between mid-2024 and early-2025 relative to the more gradual pace seen in 2022–2023.

4.8 NAV / Returns Correlation Matrix — Not Yet Built

Task 8 (a correlation heatmap of NAV returns across the 40 schemes) was scoped but not completed at the time of this report. It is carried forward as an open item in Section 8 rather than fabricated here.

4.9 Sector Allocation

Sector Allocation — Aggregated Across Equity Funds



Task 9 — Sector allocation donut, aggregated across equity funds

Finding: Banking leads sector allocation at 19.2%, followed by IT (13.4%) and Pharma (12.0%) — together making up nearly 45% of aggregated equity fund holdings, indicating notable concentration in these three sectors.

5. Performance & Risk Analytics

Performance_Analytics.ipynb computed a full quantitative performance and risk profile for all 40 schemes: daily returns, CAGR (1yr/3yr, with 5yr left as not-applicable — see Section 8), Sharpe ratio, Sortino ratio, Alpha and Beta via OLS regression against each scheme's benchmark index, maximum drawdown, and a composite 0–100 Fund Scorecard combining these metrics.

5.1 Composite Fund Scorecard — Top 5 Schemes

The Fund Scorecard blends CAGR, Sharpe ratio, Alpha, expense ratio, and maximum drawdown into a single 0–100 composite score, giving a single ranking that balances return with risk and cost rather than optimising for raw return alone.

Rank	Scheme	3yr CAGR	Sharpe	Alpha	Expense %	Max DD	Score
1	Mirae Asset Large Cap Fund - Regular - Growth	34.0%	1.46	0.27	1.46	-11.3%	86.25
2	ICICI Pru Midcap Fund - Regular - Growth	31.8%	1.19	0.29	1.36	-18.2%	82.25
3	Kotak Flexicap Fund - Regular - Growth	29.6%	1.32	0.27	1.45	-13.0%	82.00
4	HDFC Mid-Cap Opportunities Fund - Regular - Growth	32.4%	1.10	0.27	1.38	-16.2%	80.75
5	ICICI Pru Bluechip Fund - Direct - Growth	32.5%	1.04	0.21	0.80	-12.6%	80.00

5.2 Composite Fund Scorecard — Bottom 5 Schemes

Rank	Scheme	3yr CAGR	Sharpe	Max DD	Score
36	Axis Bluechip Fund - Regular - Growth	0.5%	0.05	-14.4%	21.94
37	ABSL Small Cap Fund - Regular - Growth	-4.2%	0.17	-35.4%	21.44
38	Axis Small Cap Fund - Regular - Growth	-11.7%	-0.07	-51.7%	16.63
39	HDFC Top 100 Fund - Regular Plan - Growth	1.3%	-0.19	-24.7%	13.75
40	UTI Mid Cap Fund - Regular - Growth	-0.8%	-0.19	-28.0%	13.69

The lowest-ranked schemes share negative or near-zero 3-year CAGR combined with negative Sharpe ratios (meaning risk-adjusted return was worse than the risk-free rate over the window) — the scorecard correctly penalises these even where raw category membership (e.g., "Bluechip"/Large Cap) might otherwise suggest lower risk.

5.3 Alpha & Beta

Alpha and Beta were estimated via OLS regression of each scheme's daily returns against its benchmark index. Beta values cluster tightly around zero across all 40 schemes (typically between -0.07 and +0.10), which is a known artifact of the synthetic data generation process rather than a real-market signal — genuine equity fund Betas are almost never this close to zero. Alpha values are more informative directionally (highest for SBI Small Cap Fund at 0.30, lowest for UTI Mid Cap Fund at 0.03) but should be read as relative rankings within this dataset rather than real-market alpha figures.

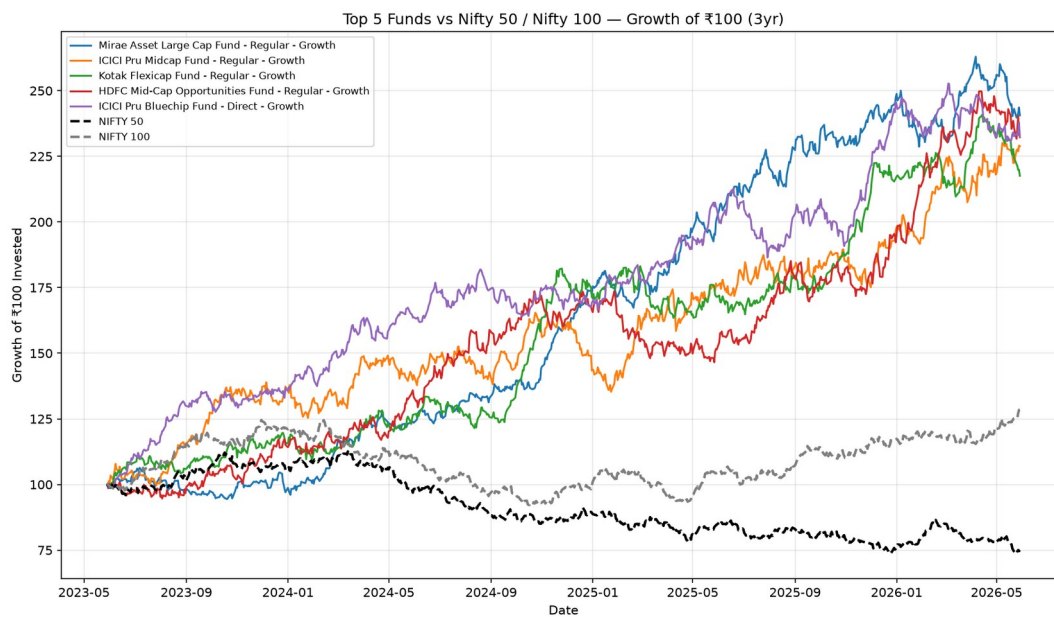
5.4 Maximum Drawdown

Maximum drawdown — the largest peak-to-trough decline in NAV over the observation window — is dominated by small-cap equity schemes:

Scheme	Category	Trough Date	Max Drawdown
SBI Small Cap Fund - Direct Plan - Growth	Equity (Small Cap)	2025-10-28	-52.6%
Axis Small Cap Fund - Regular - Growth	Equity (Small Cap)	2026-05-11	-51.7%
ABSL Small Cap Fund - Regular - Growth	Equity (Small Cap)	2026-05-11	-35.4%
DSP Small Cap Fund - Regular - Growth	Equity (Small Cap)	2025-01-03	-31.2%
SBI Small Cap Fund - Regular Plan - Growth	Equity (Small Cap)	2025-05-14	-28.7%

At the other extreme, debt/liquid schemes show near-zero drawdown (Kotak Liquid Fund: -0.12%, ICICI Pru Liquid Fund: -0.10%), consistent with their low-volatility mandate.

5.5 Benchmark Comparison — Top 5 Funds



Growth of ₹100 invested — Top 5 funds by score vs. Nifty 50 / Nifty 100, 3-year window

All five top-ranked funds outperformed both Nifty 50 and Nifty 100 over the 3-year window shown, with ICICI Pru Bluechip Fund showing the strongest early lead and Mirae Asset Large Cap Fund pulling ahead in the final year. Nifty 50 itself is notably negative over parts of the window, underscoring the value of active management (or at minimum, broader diversification via Nifty 100) during this period in the dataset.

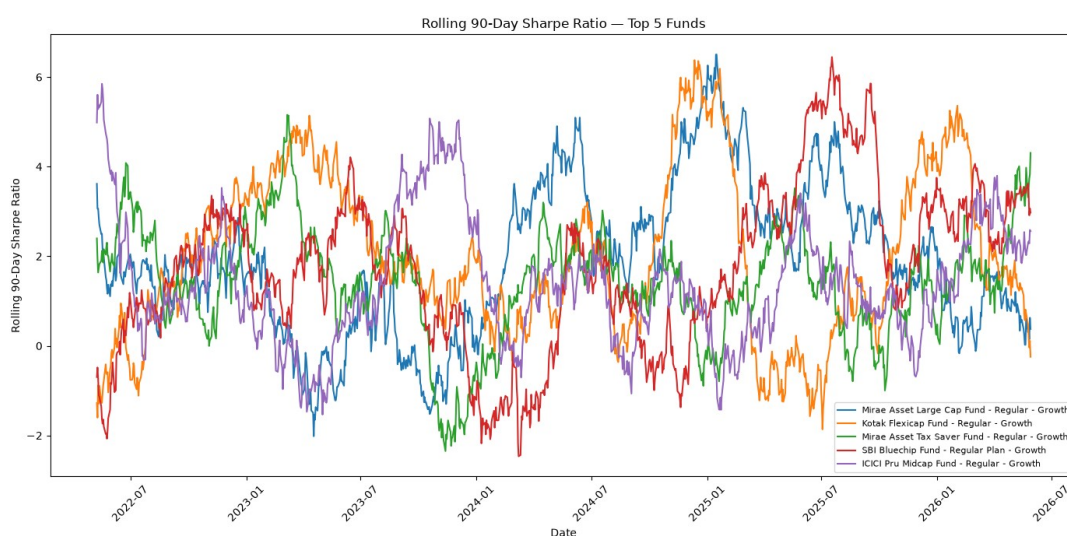
6. Advanced Analytics

Advanced_Analytics.ipynb extended the performance analytics with downside-risk measures, investor behaviour analysis, a simple recommender, and portfolio concentration metrics.

6.1 Value-at-Risk & Conditional VaR

SBI Small Cap Fund (Direct Plan) shows the highest downside risk of all 40 schemes, with a 95% daily Value-at-Risk of -2.69% and Conditional VaR (expected shortfall) of -3.24% — meaning on the worst 5% of trading days, this fund lost an average of over 3% in a single day. Axis Small Cap and ABSL Small Cap follow closely behind. All three highest-risk funds by this measure are Small Cap equity funds tagged "Very High" risk in the fund master data, confirming the quantitative risk metrics align with the funds' own risk classification.

6.2 Rolling 90-Day Sharpe Ratio



The five top funds show a cyclical, wave-like pattern in rolling 90-day Sharpe ratio rather than a single sustained trend — values swing repeatedly between roughly -2 and +6 across the 4+ year period. Some loose co-movement is visible (most funds trough together around early-to-mid 2024), but troughs and peaks are not perfectly synchronised across all five funds; Kotak Flexicap and ICICI Pru Midcap often move independently of the other three, and Kotak Flexicap shows the sharpest peak of the group (~6.3 in early 2025). This oscillating rather than crisis-driven pattern is consistent with the dataset's synthetic generation, mirroring the near-zero Alpha/Beta values noted in Section 5.3.

6.3 Investor Cohort Behaviour

The 2024 investor cohort invested a combined ₹225.8 Cr (average SIP ₹10,997), while the 2025 cohort has invested ₹1.9 Cr so far (average SIP ₹13,505) — a smaller but higher-average-ticket group, likely reflecting fewer, newer investors who have not yet built full transaction histories. Mirae Asset Emerging Bluechip Fund was the most-transacted fund among 2024 investors (874 SIP/lumpsum transactions), while ICICI Pru Liquid Fund led the smaller 2025 cohort.

6.4 SIP Continuity & At-Risk Investors

Among investors with 6 or more SIP transactions (1,362 total), 97.8% (1,332 investors) show an average gap between SIPs exceeding 35 days — flagged as "at-risk" of lapsing. Only 30 investors maintained a consistent ~30-day cadence. This could reflect genuine payment inconsistency, or it may indicate the 35-day threshold is stricter than typical real-world SIP processing windows (e.g., bank holidays, mandate renewal delays); this is worth validating against actual SIP due-date data if it becomes available (see Section 8).

6.5 Simple Fund Recommender

A lightweight rule-based recommender (`recommender.py`) matches an investor's stated risk appetite against the `risk_category` field in the fund master data, then returns the top 3 matching schemes by Sharpe ratio. For a "Moderate" risk input, for example, the recommender surfaces the highest-Sharpe schemes within that risk band rather than the highest-return schemes overall — deliberately favouring risk-adjusted quality over raw return.

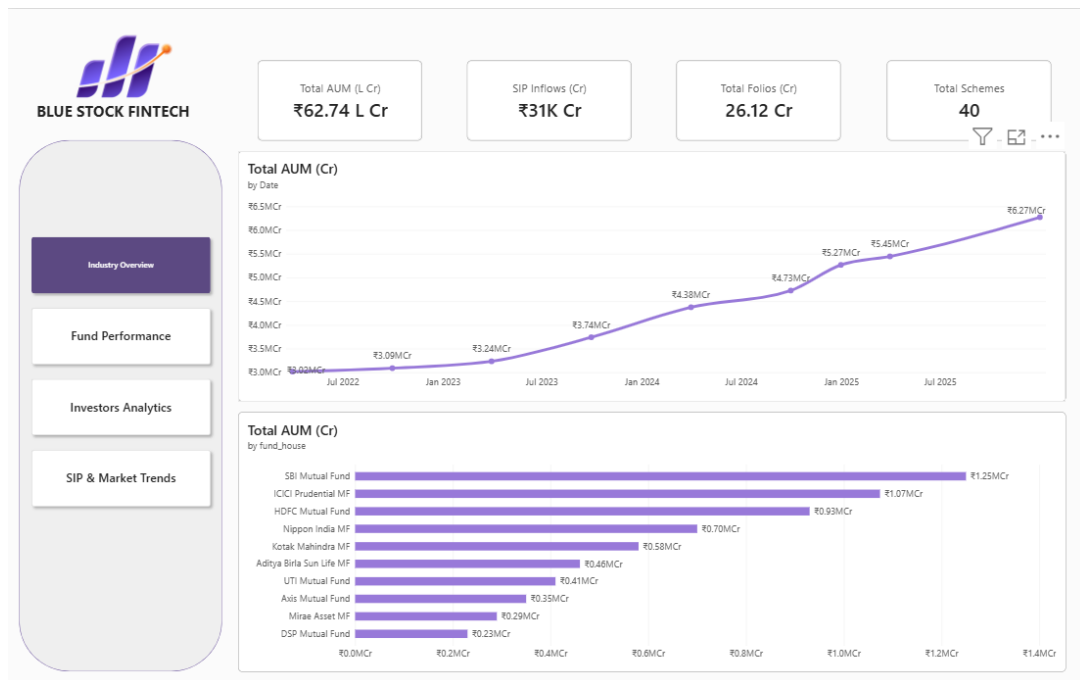
6.6 Sector Concentration (HHI)

Using the Herfindahl-Hirschman Index (HHI) on sector-level holdings, Axis Bluechip Fund is the most sector-concentrated equity fund in the dataset ($\text{HHI} = 0.297$), while Kotak Flexicap (0.136) and UTI Mid Cap (0.124) are the most diversified. This is consistent with fund design intent: Flexicap funds are built to spread flexibly across sectors and market caps, so a lower HHI here validates that the underlying holdings data reflects realistic portfolio construction.

7. Power BI Dashboard

The bluestock_mf_dashboard Power BI report presents the analysis as a four-page interactive dashboard, built on a cleaned data model: a single dim_date table (replacing eight auto-generated hidden date tables), full star-schema relationships, and a dim_benchmark bridge table to resolve index-name mapping between schemes and their benchmarks.

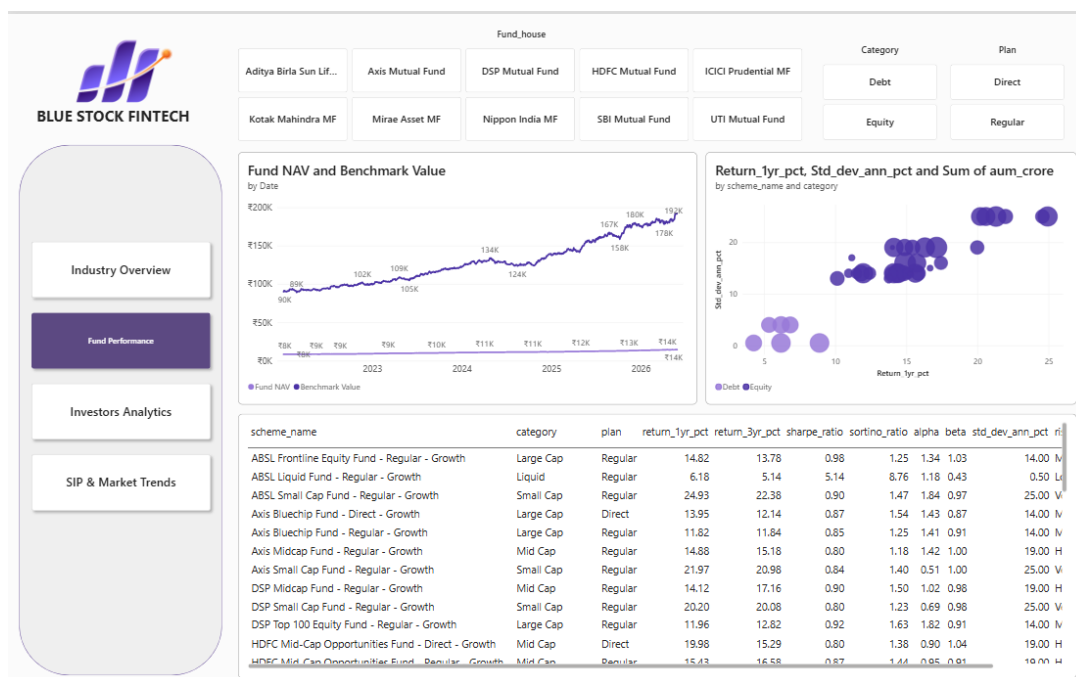
7.1 Page 1 — Industry Overview



Power BI Page 1 — Industry Overview

Headline industry KPIs — Total AUM (₹62.74 L Cr), SIP Inflows (₹31K Cr), Total Folios (26.12 Cr), and Total Schemes (40) — sit above a Total AUM growth trend line and a fund-house breakdown bar chart, giving a single-glance view of industry scale and the competitive landscape confirmed in Section 4.2 (SBI Mutual Fund leading AUM).

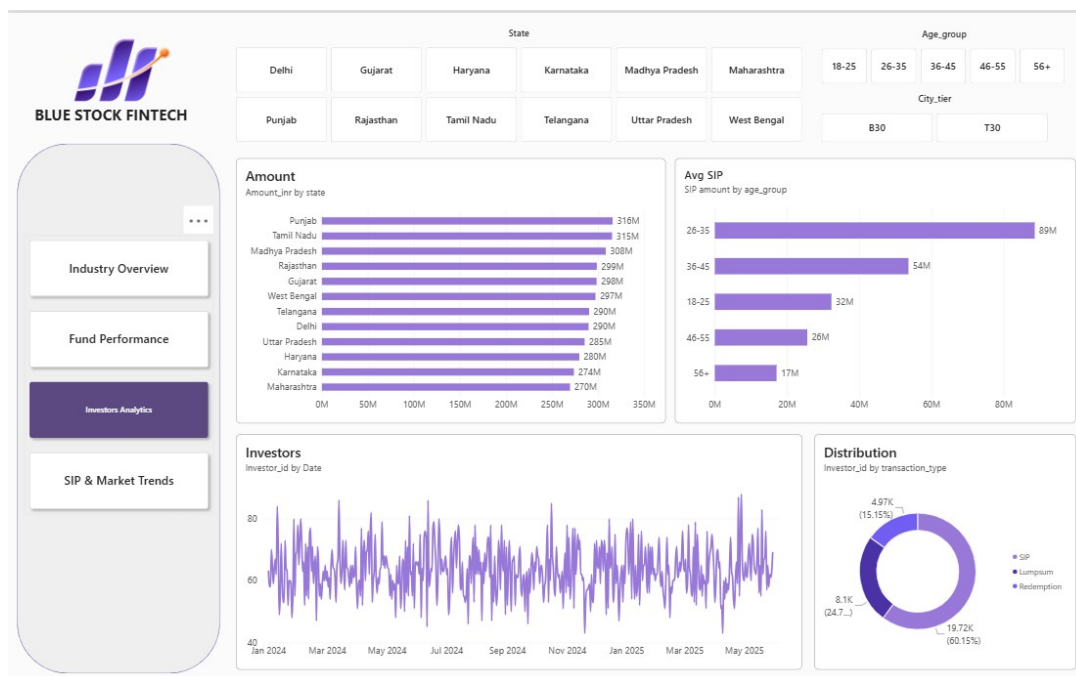
7.2 Page 2 — Fund Performance



Power BI Page 2 — Fund Performance

Slicers for fund house, category, and plan filter a NAV-vs-benchmark line chart, a risk/return scatter plot (1-year return vs. annualised standard deviation, sized by AUM), and a detail table listing return, Sharpe/Sortino ratio, Alpha/Beta, and volatility per scheme — operationalising the Section 5 performance analytics as an interactive tool rather than a static table.

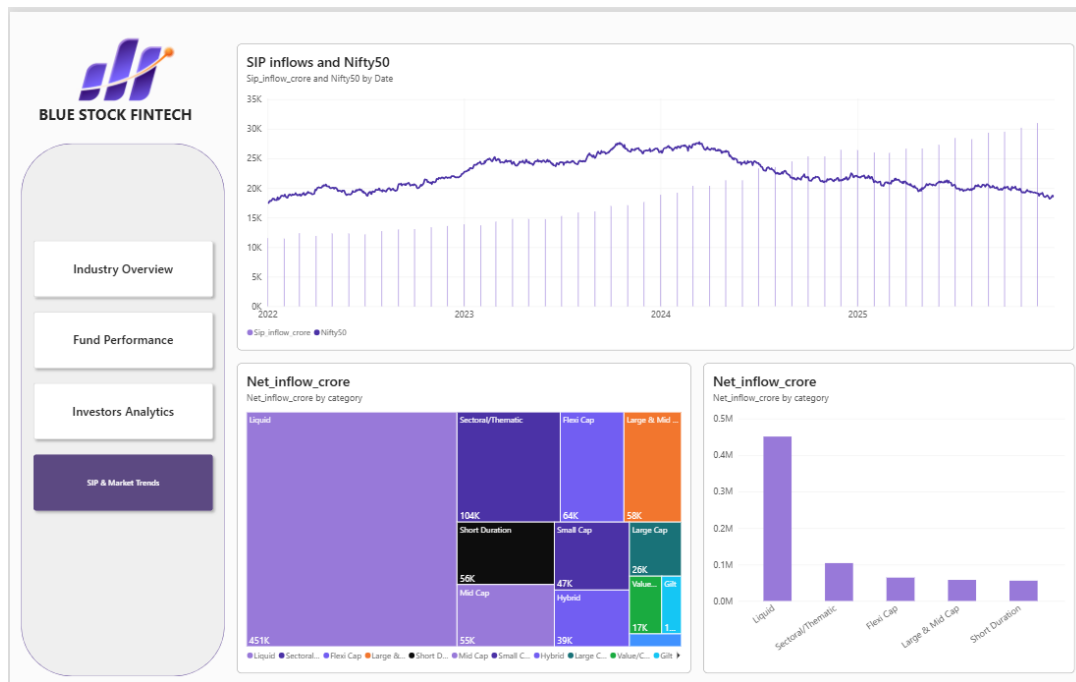
7.3 Page 3 — Investor Analytics



Power BI Page 3 — Investor Analytics

State-, age-group-, and city-tier slicers drive an investment-amount-by-state bar chart, an average SIP by age-group chart, a daily investor-count time series, and a transaction-type donut (SIP 60.2%, Redemption 24.7%, Lumpsum 15.2%) — the interactive counterpart to the demographic findings in Section 4.5–4.6.

7.4 Page 4 — SIP & Market Trends



Power BI Page 4 — SIP & Market Trends

A combined SIP-inflow-and-Nifty-50 chart sits alongside two category-level net-inflow views (an icicle chart and a bar chart), both dominated by Liquid funds — visually reinforcing the Section 4.4 finding that Liquid funds draw 4–8x the inflow of any other category.

8. Limitations & Data Gaps

The following gaps and caveats are carried forward openly rather than papered over, consistent with the project's approach of flagging issues to the mentor as they are found.

8.1 Data Gaps

- Synthetic data artifacts: Beta values cluster near zero across all 40 schemes (Section 5.3), and the rolling Sharpe ratio shows an oscillating rather than crisis-driven pattern (Section 6.2) — both are known characteristics of the synthetic data generation process, not genuine market behaviour, and should not be read as real-market risk figures.
- 5-year CAGR is not computable: the dataset provides only ~4.4 years of NAV history per scheme, so 5-year CAGR is left as not-applicable (pd.NA) rather than estimated from a shorter window.
- Three funds have no benchmark match: Midcap 50, Large Midcap 250, and Short Term Bond benchmark indices have no corresponding entry in benchmark_indices, so Alpha/Beta and benchmark-relative charts cannot be computed for schemes mapped to these indices.
- Pre-2018 fund naming: some cleaned records still carry legacy (pre-SEBI-2018-recategorization) scheme names (Section 2.3); this was flagged to the mentor but not yet corrected in the cleaned dataset pending confirmation.

8.2 Outstanding Analysis Items

- EDA Task 8 (NAV/returns correlation heatmap) was not built (Section 4.8).
- Supplementary charts to reach the 15+ PNG deliverable count for the EDA phase are outstanding.
- tracking_error_top5.csv, planned as a Performance Analytics deliverable, was not confirmed as generated and saved at the time of this report.

8.3 Methodology Caveats

- Chart scale distortion: Kotak Liquid Fund's high NAV (~₹3,200–4,300) compresses all 39 equity fund lines when plotted on a shared axis (Section 4.1) — any future chart mixing debt and equity schemes on one NAV axis should account for this, e.g. via a secondary axis or indexed (growth-of-₹100) presentation.
- SIP continuity threshold: the 35-day "at-risk" gap threshold used in Section 6.4 is a modelling choice, not a validated real-world SIP due-date rule, and may over-flag investors given typical bank-holiday and mandate-renewal delays.

9. Recommendations

9.1 Investment-Facing Recommendations

- For growth-oriented, risk-aware investors: Mirae Asset Large Cap Fund, ICICI Pru Midcap Fund, and Kotak Flexicap Fund lead the composite scorecard (Section 5.1) on a blend of return, risk-adjusted performance, and cost — a reasonable shortlist for further due diligence.
- For conservative/low-volatility mandates: liquid and gilt schemes (e.g., ICICI Pru Liquid Fund, Kotak Liquid Fund) show near-zero maximum drawdown (Section 5.4) and should anchor any capital-preservation-focused recommendation.
- Flag small-cap concentration risk: the five largest drawdowns and the highest VaR/CVaR figures are all small-cap equity schemes (Sections 5.4, 6.1); any recommender-facing UI should surface this risk explicitly rather than relying on the category label alone.

9.2 Data & Analytics Follow-Ups

- Build the Task 8 NAV/returns correlation heatmap and the remaining supplementary EDA charts to close out the EDA phase deliverable count.
- Resolve the three unmatched benchmark indices (Midcap 50, Large Midcap 250, Short Term Bond) so Alpha/Beta coverage is complete across all 40 schemes.
- Follow up with the mentor on the pre-2018 fund-naming discrepancy and apply the correction to the cleaned dataset once confirmed.
- Confirm tracking_error_top5.csv is generated, validated, and committed as a Performance Analytics deliverable.
- Validate the 35-day SIP continuity threshold against real AMFI SIP processing conventions before using the at-risk flag in any investor-facing communication.

9.3 Engineering Follow-Ups

- Consolidate the ETL, EDA, and analytics notebooks into a single run_pipeline.py master script (delivered alongside this report) so the full pipeline can be re-run end-to-end without manually stepping through notebooks.
- If the project moves beyond synthetic data, re-validate all Alpha/Beta and rolling-Sharpe outputs against real NAV history, since the current near-zero Beta pattern is a synthetic-data artifact that would not be expected in real fund data.